

Local Economic Impacts of Wind Power Deployment in Denmark

Data replication package

READ-ME

1 Content of the data replication package

- Data
 - "*Data Sources and Panel*": this excel file contains all original data we downloaded as well as the modified GIS data to build the final panel. Detailed explanations are provided in Section 3.
 - "*Code_Data*" code used for the creation of the panel (do file).
 - "*FinalPanel*": final panel (in dta format)
- Estimations
 - The remaining folders include the code and produced for each the content. Each folder will contain Stata files with the estimation codes and, if appropriate, a folder to save the outputs
 - The panel used for all analyses, named: "FinalPanel.dta"
 - The code for the main estimations, "GPS_OLS_Estimations.do" and the Output folder to save the outputs

2 Data Availability

All data used in the analysis are publicly available for download without constraints.

3 Building of the panel

The excel-file "Data Sources and Panel.xls" contains all originally downloaded and modified data (from QGIS) that are used to built the panel. It is organised in sections in the following order from right to left:

- The originally downloaded data files for the municipal budget, personal income, employment, wind turbine data, average wind density, electricity price and control variables. Anything here is directly downloaded from one of the original data sources.
- The intermediate wind turbine and wind density data, after modification with QGIS.

The final panel has a municipality-year format. We apply several steps to the original data to obtain the final data set.

3.1 Original data

Municipal GIS Data

The data on the geographical administrative borders come from the Digital Atlas of the Danish Historical-Administrative Geography (DIGDAG), which is an open-source website provided by the Danish Ministry of Higher Education and Science.^a

Income, Municipal Budget and Sectoral Employment

The original data for the income categories, municipal budget and sectoral employment come from Statistics Denmark. They are saved in the "Datasheets-Original" segment under "Datasheets Statistic Denmark". The data are at the municipal-level.

Master Data Register of Wind Turbines

The Master Data Register of Wind Turbines is a national database which contains all Danish power producing wind turbines that are larger than 6kW. The database contains technical data, X-Y coordinates, power output data, as well as the commission and decommission dates of all wind turbines from 1977-2021. The register separates them into two groups: one sheet for all operating turbines and one sheet for the decommissioned turbines. The X-Y coordinates in the register are a combination of coordinates from GeoDenmark data.^b

Electricity Price

To determine the electricity price received by wind power producers, we use the electricity price data from Dansk Energi^c for the time period 1985-1999 (before the electricity market liberalisation). For the time period 2000-2010, we compute the weighted electricity price, based on the hourly wind power production and hourly prices from Energinet,^d as follows:

$$\sum_h (price_h \times production_h) / \sum_h production_h$$

New Revenues

We compute the revenues that turbine owners receive in the first year of connection to the grid, what we call "New Revenues". The computation takes into account the electricity price and the support policy when the turbine was built. The detailed information on support policies are provided by EnergiStyrelsen (2011).^e The Master Data Register on Wind Turbines provides the power production of each turbine in each year. Several support scheme were used in Denmark: price premium, fixed feed-in tariff, and price premium with an upper bound on the sum of electricity price and support (the support scheme depends on the year of commissioning of the turbines). On top of these, an allowance of 0.3 cEuro/kWh is paid to wind power producers for balancing costs for the first twenty years. The *New Revenues* are hence computed as:

$$PowerProduction_{y0} * (ElectricityPrice_{y0} + Support_{y0} + Allowance_{y0})$$

with all variables taken in $y0$, the year of commissioning

^a<https://www.digdag.dk/>

^b<https://ens.dk/en/our-services/statistics-data-key-figures-and-energy-maps/overview-energy-sector>

^c<https://www.danskenergi.dk/>

^d<https://en.energinet.dk/>

^eVindmølleindustrien som historisk flagskib", J.nr. Ref. PCJ/HW/FGN 24. maj 2011 (English version: "The history of Danish support for wind power")

Controls

Population and Unemployment

The original data for the population and unemployment come from Statistics Denmark. They are saved in the "Datasheets Statistic Denmark" sheet in the "Datasheets-Original" section. We further compute the unemployment rate as the number of unemployed people divided by the sum of employed and unemployed people in the municipality.

Available agricultural land surface

We compute a proxy for the available agricultural land surface, using data on agricultural land surface at the county level for our time period of interest and the data on agricultural land surface at the municipal level from a 1999 census. Both data sets are obtained from Statistics Denmark and saved in the "Datasheets Statistic Denmark" sheet in the "Datasheets-Original" section. We assume that the share of agricultural land surface of each municipality in the overall agricultural land surface of the corresponding county remains constant. We compute the proxy for agricultural land surface as the multiplication of the municipality's share in its county's agricultural land surface (assumed constant) by the yearly data of agricultural land surface at the county level.

Average Wind Density

We obtain data on the average wind density from the "Global Wind Atlas (GWA)".^f We download the average wind density data for Denmark and save them in a Geo-tiff format. We import this file into QGIS and select the same coordinate system (KBS) for the TIFF-Data and our municipal shape files. We finally use the tool "zonal statistics" and compute the average wind density for each municipality. The result of this process is saved in the sheet "WindPowerDensity".

Stock of operating Wind Turbines

We obtain the stock of operating wind turbines from the Master Data Register. We compute the stock as the difference between the cumulative sum of commissioned and decommissioned wind turbines.

3.2 Work to prepare the panel

For each data source, we reshape the data into a municipality-year format, we assign a number and manually harmonize the municipality names, as they do not exactly match across data sources (ex: some data sources use "Kobenhavn" while others use "Copenhagen"). We manually correct spelling errors in the municipalities names (the names of Danish municipalities include special letters - like ø, å, æ- which are not always well handled by QGIS, Stata and Excel). These changes are saved in the TEMP-Folder of the data section.

The municipality of Christiansø is outside of the geographical area of the study. It has so little data that we have not kept it in the estimation.

Before we obtain the final panel, we apply three edits.

- We identify and merge the 21 urbanised small municipalities that belong to the greater county of Copenhagen (see do file).
- We merge Marstal, Ærøskøbing and Ærø, as they represent the same municipality but its name changed over time (e.g Ærø and Marstal until 2005, Ærøskøbing from 2005 onwards).
- The island of Bornholm consists of the following municipalities: Aakrikreby, RÅžnne, NexÅž, Hasle and Allinge-Gudhjem. For these five municipalities, the data are sometime not available after 2002

^fThe GWA application is owned and maintained by the Technical University of Denmark: <https://globalwindatlas.info/>.

(depending on the data source). We drop these five municipalities and instead use the county data (Bornholm) for which we have the complete data until 2006.

For each of these edits, we generate, use and store temporary files. These files are saved in the TEMP-Folder with the corresponding prefix.[§]

Then, all the previously mentioned data-sheets at the municipal-level are merged into the final panel for the estimations.

We change the data on the municipal budget, income and the new revenues from Danish Krone (DKK) to Euros and generate the per-capita values for the variables of interest.

We finally obtain a balanced panel covering 251 municipalities, erase all temporary data sets (TEMP-data), and label the variables accordingly.

4 Estimation

This final panel is saved in dta format. The design of the tables and figures was modified by hand using the Stata Graph Editor for the graphs and LaTeX for the tables, but the numbers and information included are the same. The do-files are commented.

[§]The prefix *Copenhagen_* refers to the consolidation of the greater county of Copenhagen, the prefix *Aeroskøbing_* refers the merge of Marstal, Ærøskøbing and Ærø, and the prefix *Bornholm_* refers to the five municipalities of the island of Bornholm.